

CRR AS A TEMPORAL ARCHITECTURE FOR THERMODYNAMIC AND MORTAL COMPUTATION

*Endogenous Clocks, Physical Memory, Offline Regeneration, World Models, and
Distributed Mortal AI*

A formal exploratory synthesis of Coherence-Rupture-Regeneration with thermodynamic computing, stochastic thermodynamics, mortal computation, active inference, sleep/replay, phenomenology, and distributed intelligence

Prepared 30 August 2026

Status: exploratory theoretical and engineering proposal; not peer reviewed.

Scope and epistemic status

The canonical CRR equations are presented as the current object of study, including the unresolved public convention between Fisher-Rao arc-length and Fisher squared-speed clocks. All architecture, metrics, and protocols developed after the canonical section are proposed extensions. “Dreaming” denotes offline internally generated world-model activity and does not imply phenomenal consciousness.

Abstract

Thermodynamic computing and mortal computation challenge the conventional separation between software and substrate. Contemporary thermodynamic hardware already exploits physical fluctuations for probabilistic sampling and linear-algebra primitives, while new nonlinear thermodynamic neural networks extend this idea to out-of-equilibrium universal approximation. Mortal computation proposes that low-power intelligence may require learning rules that exploit the unique physical properties of each hardware instance, making learned computation non-portable and therefore literally mortal. This paper asks whether Coherence-Rupture-Regeneration (CRR) can supply a missing temporal architecture for such systems. CRR is interpreted not as a replacement for stochastic thermodynamics, the Free Energy Principle (FEP), or world-model learning, but as a candidate endogenous lifecycle controller: accumulate distinguishable adaptive change; commit or reorganize at a data-defined stopping event; retain selected physical history; and reconstruct from that archive with controllable breadth. A direct Gibbs implementation of the CRR regeneration kernel is derived, together with a substrate-specific mortality index, coherence-per-dissipation measure, regeneration value, dream-gain-per-joule measure, and distributed phase-order metric. A wake-offline-regeneration architecture is proposed in which online interaction accumulates processual coherence and offline phases use temperature/noise modulation to perform replay, consolidation, broad generative recombination, and world-model refinement. The framework is connected to Hinton’s mortal computation and Forward-Forward work, Ororbia and Friston’s FEP treatment, current thermodynamic computers, stochastic halting-time thermodynamics, hippocampal replay, generative memory, Dreamer-style world models, Husserlian temporal phenomenology, and Watson-Levin multiscale intelligence. Eight empirical protocols and explicit falsifiers are specified. The main claim is modest but testable: CRR may be useful if process-defined stopping, substrate-bound archive dependence, and history-weighted offline reconstruction improve adaptive performance or thermodynamic efficiency beyond fixed-time and ordinary replay baselines.

Keywords: thermodynamic computing; mortal computation; Coherence-Rupture-Regeneration; stochastic thermodynamics; Free Energy Principle; physical AI; world models; sleep; replay; continual learning; distributed intelligence; phenomenology; neuromorphic computation.

Contents

1. Introduction: computation that has a lifetime
 2. Canonical CRR and the clock convention
 3. Current frontier: thermodynamic and mortal computation
 4. The CRR-native thermodynamic computer
 5. Rupture as endogenous halting and physical commitment
 6. Physical archives and an empirical definition of mortality
 7. CRR, FEP, and thermodynamic world-model sampling
 8. Wake, sleep, replay, and an AI “dreaming” analogue
 9. Continual learning and the stability-plasticity problem
 10. Distributed mortal AI and the relational weave
 11. Phenomenology and process philosophy: what is and is not licensed
 12. Engineering metrics and experimental protocols
 13. Predictions, falsifiers, and failure modes
 14. Research programme and conclusion
- References
- Appendix A. Mathematical notation and implementation notes

1. Introduction: computation that has a lifetime

Most digital computation is designed to make logical function independent of physical history. If a processor fails, the same program and parameters can be loaded on another sufficiently equivalent device. Thermodynamic and mortal computation deliberately relax that ideal. Thermodynamic computing treats thermal fluctuations, relaxation, and nonequilibrium trajectories as computational resources. Mortal computation goes further: a learner adapts to the idiosyncratic analogue properties of one physical substrate, so its learned state is not fully portable and dies with the machine (Hinton, 2022; Ororbis & Friston, 2023).

This shift exposes a temporal problem that conventional computer architecture can often ignore. A physical learner must decide when a transient computation has done enough work, when to commit to an output, when to reset volatile degrees of freedom, what persistent physical traces to preserve, when to reduce sensory coupling, and how to reorganize its internal model without erasing useful history. These are lifecycle questions rather than objective-function questions.

CRR is potentially useful precisely because it is a grammar of lifecycle. Its three operations - coherence, rupture, and regeneration - can be read as an endogenous process clock, an event boundary, and a history-conditioned reconstruction operator. The central proposal of this paper is therefore not “CRR explains thermodynamics.” Stochastic thermodynamics already supplies the relevant physics. The proposal is that CRR may serve as a compact temporal controller for thermodynamic intelligence.

Core engineering thesis

Thermodynamics supplies the substrate and energetic accounting. FEP or another objective supplies adaptive valuation. World models supply counterfactual futures. Mortal computation supplies substrate-specific learning. CRR is proposed to supply when a regime has accumulated enough change to commit, what physical history survives, and how broadly that history is sampled during reconstruction.

2. Canonical CRR and the clock convention

The public CRR formulation is deliberately parsimonious: accumulated coherence C , a rupture event δ when capacity is exhausted, and a regeneration functional R that weights an archive ϕ by an exponential kernel governed by Ω (Sabine, 2026a,b). There is, however, an unresolved versioning issue. The public website describes L as Fisher-Rao speed and C as accumulated arc length, whereas a canonical repository iteration used Fisher squared speed. These are different functionals. Rather than silently choose one, this paper carries them as competing candidate clocks.

2.1 Candidate clock A: Fisher-Rao arc length

$$v_{FR}(t) = \sqrt{\dot{\theta}(t)^T G(\theta(t)) \dot{\theta}(t)} \quad (1)$$

$$C_{\ell,n}(t) = \int_{t_n}^t v_{FR}(\tau) d\tau \quad (2)$$

This is the form currently described on temporalgrammar.ai: C is cumulative path length in information geometry. It is invariant to smooth reparameterization of the statistical coordinates, provided the same probability model is used.

2.2 Candidate clock B: Fisher action

$$C_{A,n}(t) = \int_{t_n}^t v_{FR}(\tau)^2 d\tau = \int_{t_n}^t \dot{\theta}^T G(\theta) \dot{\theta} d\tau \quad (3)$$

This squared-speed form behaves like an informational action: rapid distributional changes count disproportionately. It is not equivalent to arc length and generally produces different stopping predictions. In all proposed experiments both clocks should be computed from the same data and compared out of sample.

2.3 Rupture

$$\delta_n = \delta(t - t_n), \quad t_{n+1} = \inf \{t > t_n : C_n(t) \Omega_n \geq 1\} \quad (4)$$

Equation (4) is the strong CRR threshold hypothesis. For engineering, the event itself should be defined independently - readout, phase transition, synchronization event, reconfiguration, task completion, or stochastic stopping - and only then should the $C\Omega$ rule be tested. If it fails, CRR can still be evaluated as a heuristic clock with a learned hazard.

2.4 Regeneration

$$R_a(t) = \int_{-\infty}^t \phi_a(\tau) \exp\left(\frac{C_a(\tau)}{\Omega}\right) \Theta(t - \tau) d\tau \quad (5)$$

$$P_{\text{CRR}}(a | t) = \frac{R_a(t)}{\sum_b R_b(t)} \quad (6)$$

Here a indexes candidate reconstructed states, policies, attractors, memories, or model configurations. The archive ϕ is the physically available material from which reconstruction can occur. Low Ω makes high-coherence history dominate; high Ω broadens access to the archive. The original CRR paper explicitly separates the cleared accumulator from the distributed archive and links such substrate-bound persistence to mortal computation (Sabine, 2026b, Section 10.1).

Symbol	CRR role	Engineering interpretation
θ, p	Statistical state	Distribution over physical or latent states of the device
G	Fisher metric	Geometry used to quantify distinguishable distributional motion
v_{FR}	Fisher-Rao speed	Instantaneous information-geometric process rate
C_1 / C_A	Candidate coherence clocks	Accumulated arc length / accumulated squared-speed action
Ω	Reconstruction/capacity parameter	Calibrated breadth or effective temperature-like control; strong geometric interpretation tested separately
δ	Rupture	Commit/readout/reconfiguration/stopping event
ϕ or Φ	Archive	Slow physical state: conductances, hysteresis, persistent chemistry, device-specific latent memory
R	Regeneration	History-conditioned reconstruction or sampling of the next regime

3. Current frontier: thermodynamic and mortal computation

Thermodynamic computing is no longer only a thought experiment. Melanson and colleagues demonstrated an eight-cell stochastic processing unit built from coupled RLC circuits that samples multivariate Gaussian distributions and performs matrix inversion by exploiting physical thermal/noise-driven dynamics (Melanson et al., 2025). Whitelam and Casert subsequently developed nonlinear thermodynamic neurons with non-quadratic potentials and showed that networks of them can perform arbitrary nonlinear calculations in or out of equilibrium, extending thermodynamic computation beyond equilibrium Boltzmann machines (Whitelam & Casert, 2026).

Stochastic thermodynamics has independently moved toward realistic computational features. Manzano, Kardes, Roldan, and Wolpert derive fluctuation relations and dissipation bounds for computations with stochastic stopping

times, unidirectional transitions, restricted initial conditions, and absolute irreversibility - precisely the features expected in physical adaptive hardware (Manzano et al., 2024). Wolpert and Korbel review the broader promise and subtleties of applying stochastic thermodynamics to computation (Wolpert & Korbel, 2024).

Mortal computation introduces a complementary constraint. Hinton argues that low-power analogue learners may need to exploit the unique physical properties of a particular hardware instance; their internal learned parameters therefore cannot simply be copied to another nominally similar device. What can be transmitted is behaviour or knowledge through distillation rather than direct state transplantation (Hinton, 2022). Ororbia and Friston formalize this research direction using Markov blankets, FEP, circular causality, and embodied biomimetic computation (Ororbia & Friston, 2023).

Programme	What it already supplies	Possible CRR niche
Thermodynamic computing	Physical sampling, relaxation, nonlinear stochastic computation	Endogenous process clock and lifecycle switching
Stochastic thermodynamics	Dissipation, entropy production, stochastic stopping-time bounds	Candidate state-dependent stopping variable and cycle segmentation
Mortal computation	Substrate-specific learning; non-portable internal state	Formal archive dependence and measurable mortality
FEP / active inference	Adaptive inference/action objective and Markov-blanket individuation	Event timing and history-conditioned regime reconstruction
World models	Counterfactual rollout and imagination from learned dynamics	Offline phase control and archive-conditioned replay
Sleep/replay models	Memory consolidation, interference control, generative reconstruction	A tunable reconstruction breadth Ω and endogenous wake/sleep trigger

4. The CRR-native thermodynamic computer

Consider a thermodynamic computational substrate with physical state z , control parameters λ , mobility matrix M , potential U , and thermal/noise forcing. A generic overdamped Langevin model is

$$dz_t = -M \nabla_z U(z_t; \lambda) dt + \sqrt{2k_B T M} dW_t \quad (7)$$

The stationary or transient distribution $p(z, t)$ is computationally meaningful; thermal fluctuations are resources rather than errors. CRR adds a process clock computed from the changing distribution and a persistent archive Φ stored in slower physical degrees of freedom.

4.1 Exact Gibbs correspondence for regeneration

Suppose a thermodynamic sampler selects among candidate reconstructed states a according to a Gibbs law,

$$P(a) = \frac{\exp[-E_a/(k_B T)]}{Z} \quad (8)$$

Choose an energy landscape

$$E_a = -\alpha C_a - k_B T \ln \phi_a \quad (9)$$

Then

$$P(a) \propto \phi_a \exp\left(\frac{\alpha C_a}{k_B T}\right) \quad (10)$$

$$\Omega_{\text{impl}} = \frac{k_B T}{\alpha} \implies P(a) \propto \phi_a e^{C_a / \Omega_{\text{impl}}} \quad (11)$$

Key engineering result

Equation (11) is an implementation equivalence, not a claim that abstract CRR Ω literally is thermodynamic temperature. It shows that a physical Gibbs sampler can realize the CRR reconstruction law if energy wells encode historical coherence and archive mass. This gives CRR a direct path to hardware implementation.

Whitelam and Casert's nonlinear thermodynamic neurons matter because they move beyond equilibrium Gaussian sampling: a CRR machine need not wait for full equilibration and could use trained Langevin trajectories as computational regimes (Whitelam & Casert, 2026).

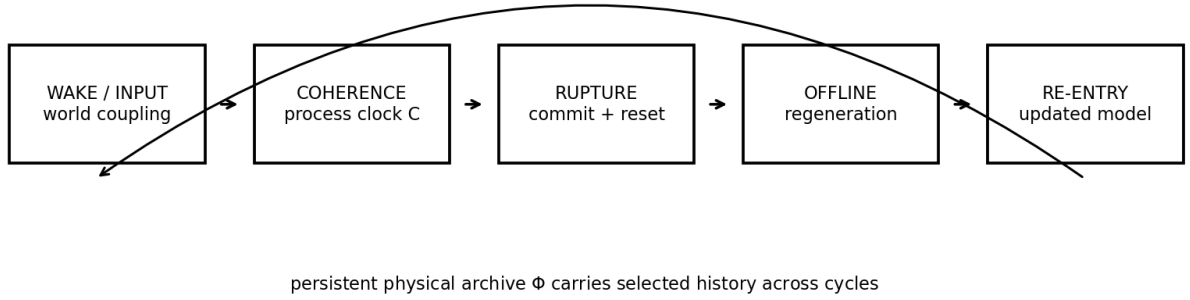


Figure 1. Proposed CRR-native lifecycle: online coupling accumulates a process clock; a stopping/commit event clears volatile state; a persistent archive drives offline regeneration before re-entry.

5. Rupture as endogenous halting and physical commitment

Digital systems often stop because an external program counter or loop condition says so. Physical stochastic computers can instead have uncertain completion times. This is a natural point of contact between CRR and stochastic thermodynamics of stopping times (Manzano et al., 2024).

The strongest CRR hypothesis is the threshold in equation (4). A safer engineering formulation replaces a hard threshold with a hazard function,

$$\Pr(\delta \text{ in } [t, t + dt] \mid \mathcal{H}_t) = h(C(t), \Omega, \Sigma(t), u(t)) dt \quad (12)$$

where Σ may include entropy production, uncertainty, task loss, or other measurable device variables. The empirical question is whether C contains predictive information about stopping that is not already captured by wall-clock time, instantaneous loss, or energy dissipation.

A minimal model-comparison experiment should compare:

- fixed-time stopping: $t = \text{constant}$;
- loss threshold: stop when task loss crosses a criterion;
- energy threshold: stop after fixed dissipated work;
- CRR arc-length clock C_l ;
- CRR action clock C_A ;
- a flexible learned hazard containing all variables.

CRR earns an engineering role only if one of its clocks predicts useful commit times out of sample with lower complexity or better energy/performance trade-offs than these baselines.

6. Physical archives and an empirical definition of mortality

For a conventional computer, the goal is approximately to make function independent of microscopic substrate history. For a mortal computer, the learned computation exploits device-specific analogue properties. CRR sharpens this by separating fast volatile state from a slower archive. The archive can be implemented by memristive conductances, hysteretic material states, slow bias variables, structural plasticity, chemical concentrations, device ageing, local oscillator phases, or other history-bearing physical variables.

$$\Phi_{n+1} = \mathcal{U}(\Phi_n, \delta_n, y_n, \xi_n) \quad (13)$$

Here $\mathcal{U}$ is a physical archive-update map, y_n the committed output, and ξ_n local stochastic/physical perturbations. At the next regime, volatile state is reconstructed conditionally on Φ rather than loaded from a perfectly portable digital checkpoint.

6.1 A mortality index

$$\mathcal{M} = \frac{I(Y_{n+1}; \Phi_n^{\text{phys}} | W_n, X_n)}{H(Y_{n+1} | W_n, X_n)} \quad (14)$$

W_n denotes all deliberately portable logical state and X_n current inputs. $\mathcal{M}$ approximately zero means the physical archive adds no predictive information once portable state is known: the machine is effectively immortal at the chosen description. $\mathcal{M}$ greater than zero means future behaviour depends on substrate history that was not captured by the portable state.

This supports a decisive transplant experiment. Train device A. Copy every portable variable W into nominally equivalent device B, but do not copy A's microscopic physical archive. If B reproduces A exactly, mortality is low. If A retains systematic behavioural capabilities or biases not recovered in B, quantify the residual with $\mathcal{M}$. Hinton's claim that knowledge may need to be distilled rather than copied becomes a measurable continuum rather than a categorical label.

6.2 Thermodynamic cost of rupture

The original CRR v01 paper proposed a fixed $k_B T \ln 2$ lower bound for every binary rupture (Sabine, 2026b, Section 10.1). For engineering, the safer statement is that logical erasure costs depend on the entropy actually removed. Landauer's familiar $k_B T \ln 2$ bound applies to erasing one unbiased bit; biased or side-information-conditioned memories can carry less entropy. The relevant measurements are therefore the entropy erased, heat dissipated, and total entropy production per cycle (Landauer, 1961; Georgescu, 2021; Manzano et al., 2024).

$$Q_{\text{erase}} \geq k_B T H_{\text{nats}}(M_{\text{erased}}) \quad (15)$$

7. CRR, FEP, and thermodynamic world-model sampling

FEP and active inference offer a natural prospective term. If candidate future state or policy a has expected free energy G_a , while CRR supplies historical support through ϕ_a and C_a , define

$$Q(a | t) \propto \phi_a(t) \exp \left[\frac{C_a(t)}{\Omega} - \gamma G_a(t) \right] \quad (16)$$

The two exponents have opposite temporal orientations. C/Ω is inherited historical support; $-\gamma G$ is prospective viability or epistemic/pragmatic value. This should be read as a proposed synthesis rather than a derivation from FEP.

For a thermodynamic implementation define the dimensionless energy

$$\mathcal{E}_a = \gamma G_a - \frac{C_a}{\Omega} - \ln \phi_a \quad (17)$$

$$Q(a) = \frac{e^{-\mathcal{E}_a}}{\sum_b e^{-\mathcal{E}_b}} \quad (18)$$

This gives a concrete research target: a stochastic physical network whose energy landscape simultaneously represents historical support and expected future cost, allowing physical fluctuations to perform approximate trajectory selection.

World-model systems already exploit imagined trajectories computationally. DreamerV3 learns a recurrent world model and trains actor and critic networks from replayed experience and imagined trajectories, demonstrating robust control across more than 150 tasks (Hafner et al., 2025). CRR’s proposed addition is not imagination itself, but lifecycle timing and the weighting of what history is allowed to generate those imagined futures.

8. Wake, sleep, replay, and an AI “dreaming” analogue

Sleep is an especially productive analogy because biology does not learn under one stationary computational regime. Sensory coupling, neuromodulation, replay statistics, and memory processing vary over structured offline phases. “AI dreaming” in this paper means internally generated world-model activity under reduced external sensory constraint. It does not imply subjective dreaming or consciousness.

There are several independent computational precedents. The classic wake-sleep algorithm separates recognition learning under external data from generative-model learning under internally generated samples (Hinton et al., 1995). Hinton’s Forward-Forward paper suggests that positive and negative phases might in principle be separated in time, with self-generated negative data processed offline, while also explicitly reporting that full temporal separation remained unresolved in his experiments (Hinton, 2022). Hobson, Hong, and Friston propose that waking updates a generative model from sensory prediction errors whereas sleep may optimize and simplify that model under altered sensory gating (Hobson et al., 2014).

Neuroscience has also become more specific. Chang and colleagues showed in 2025 that non-REM sleep contains substates that temporally segregate replay of recent versus older memories, with causal disruption of sharp-wave ripples in one substate impairing consolidation of new memory (Chang et al., 2025). Spens and Burgess model hippocampal replay as training cortical generative models that support reconstruction, imagination, semantic abstraction, and future inference (Spens & Burgess, 2024).

8.1 CRR wake-sleep control law

$$C_{\text{wake}}(t) = \int_{t_n}^t v_{\text{FR}}(\tau) d\tau \quad \text{or} \quad \int_{t_n}^t v_{\text{FR}}(\tau)^2 d\tau \quad (19)$$

During wake, external sensory precision is high and the archive is updated from real interactions. Once a stopping criterion is reached, external coupling can be reduced and an offline distribution sampled:

$$P_{\text{off}}(a) \propto \phi_a \exp\left(\frac{C_a}{\Omega_{\text{off}}} - \gamma G_a\right) \quad (20)$$

This suggests at least two offline modes. A relatively low/moderate Ω_{off} produces selective replay of high-coherence traces, useful for consolidation. A larger Ω_{off} flattens historical preference and allows remote, weak, or recombined traces to participate, useful for generative exploration and counterfactual world-model refinement.

Phase	External coupling	Ω / sampling breadth	Proposed computational function
Wake	High	Task-dependent; often narrower	Online inference, control, archive acquisition
Selective replay	Low	Lower/moderate	Consolidate salient recent or high-coherence traces
Broad generative replay	Low	Higher	Recombine distant archive content; counterfactual world modelling
Re-entry	Rising	Annealed toward online value	Test reconstructed model against

			world
--	--	--	-------

Important falsifier

If fixed replay schedules or ordinary prioritized replay consistently match or outperform CRR-triggered offline phases at equal compute and energy, then CRR has not added a useful lifecycle principle. Biological resonance is not sufficient evidence.

9. Continual learning and the stability-plasticity problem

Continual learning must avoid two symmetrical failures: catastrophic forgetting, in which new learning overwrites prior competence, and excessive stability, in which old representations block adaptation. CRR's Ω is naturally interpretable as a history-selection breadth controller. The core hypothesis is not that one fixed Ω solves continual learning, but that scheduled changes in reconstruction breadth can dynamically trade stability against plasticity.

$$w_h(\Omega) = \frac{\phi_h e^{C_h/\Omega}}{\sum_j \phi_j e^{C_j/\Omega}} \quad (21)$$

A low Ω produces a peaked replay distribution; high Ω approaches a broader archive distribution. In a CRR-style system, replay priority is therefore a physical or algorithmic consequence of coherence and Ω rather than an arbitrary buffer heuristic.

The proposed comparison should include uniform replay, recency replay, prioritized replay, reservoir sampling, conventional generative replay, and CRR weighting. Performance must be assessed not just on final accuracy but on return-to-old-context speed, forward transfer, interference, calibration, and joules spent in offline computation.

10. Distributed mortal AI and the relational weave

Mortal computation creates a transmission problem. If the full internal physical state of one device cannot be transplanted, how can a population accumulate knowledge? Hinton's answer is distillation: transmit behaviour rather than weights. CRR's boundary grammar makes the same move in process terms: what leaves one system as a committed event can become archive material for another, while internal physical history remains local. The original CRR paper states that what is archive material for one system can be a completed event from another in its boundary grammar (Sabine, 2026b, Section 2.4).

$$m_i(t) = \sum_n y_{i,n} \delta(t - t_{i,n}) \quad (22)$$

$$\Phi_i(t) = \Phi_i^{\text{self}}(t) + \sum_{j \neq i} \int_{-\infty}^t K_{ji}(t - \tau) dm_j(\tau) \quad (23)$$

Only selected commitments cross the boundary. This supports sparse event-driven communication rather than continual full-state synchronization. Such an architecture is naturally compatible with neuromorphic/event-based hardware and with populations of non-identical mortal devices.

10.1 A shared temporal field

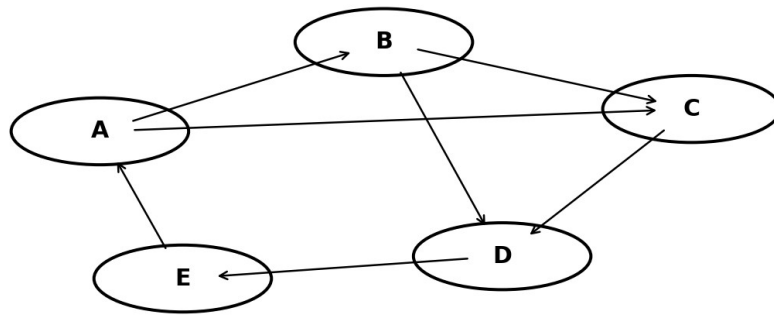
$$\theta_i(t) = 2\pi \frac{C_i(t)}{C_i^*} \bmod 2\pi \quad (24)$$

$$r(t) = \left| \frac{1}{N} \sum_{i=1}^N e^{i\theta_i(t)} \right| \quad (25)$$

The Kuramoto order parameter r measures phase alignment of process clocks. This does not prove a CRR-specific synchronisation mechanism; the original paper itself notes that coupling for CRR clocks remains incomplete (Sabine, 2026b, Section 11.2). But it creates a direct experiment: hold message content fixed while manipulating phase-

relative delivery time. If collective task performance depends on phase, timing is a genuine computational variable rather than a metaphor.

Sparse rupture events become archive material in other mortal agents



internal physical state remains local; outputs cross boundaries

Figure 2. Distributed mortal architecture. Internal substrate-specific state is not copied; sparse commitment events become delayed archive material for other agents.

Watson and Levin argue that individual and collective intelligence can be understood through relationships among competent subunits rather than a categorical divide between individual and collective agents (Watson & Levin, 2023). A CRR distributed system would add a temporal hypothesis: higher-level agency may depend partly on how local lifecycle events are coupled and inherited.

11. Phenomenology and process philosophy: what is and is not licensed

CRR is especially suggestive phenomenologically because its architecture is not a sequence of static states controlled by an external clock. It is history-laden, anticipatory, and punctuated by endogenous transitions. Husserl’s analysis of temporal consciousness distinguishes retention, primal impression, and protention: the just-past remains structurally implicated in the living present, while the present opens toward an indeterminate future horizon (Husserl, 1991; Zahavi, 2025 overview).

Phenomenological term	Possible engineering analogue	Status
Retention	Persistent archive Φ and near-history carried into current inference	Structural analogy only
Primal impression / now	Commitment boundary δ or present sensorimotor update	Structural analogy only
Protention	World-model distribution over anticipated futures	Structural analogy only
Temporal synthesis	Coupling of archive, present update, and future prediction	Potential design vocabulary

This vocabulary is useful for architecture design because it emphasizes that “present computation” is constituted by relations to retained history and anticipated possibility. It does not license an inference from such structure to phenomenal consciousness. A CRR thermodynamic agent might have retention-like memory, endogenous event time, world-model protention, sleep-like offline phases, and mortality without there being anything it is like to be that system.

Process philosophy offers a similar heuristic. The original CRR paper interprets rupture as private process becoming public datum and stresses that committed events enter the material from which later reconstructions are made (Sabine, 2026b, Sections 10.2-11). For engineering this translates cleanly: completed outputs are irreversible causal facts for the network, whereas the internal transient computation that produced them need not be externally accessible or portable.

12. Engineering metrics and experimental protocols

CRR becomes scientifically useful only if it can improve prediction, control, adaptation, or thermodynamic efficiency relative to simpler alternatives. The following metrics and protocols are designed to make that claim falsifiable.

12.1 Core metrics

$$\eta_C = \frac{\Delta C}{Q_{\text{diss}}/(k_B T)} \quad (26)$$

Coherence per thermal unit dissipated; report separately for C_l and C_A.

$$\mathcal{G}_R = \mathcal{L}_{\text{no archive}} - \mathcal{L}_{\text{archive}} \quad (27)$$

Regeneration value: held-out loss improvement due to archive-conditioned reconstruction after reset.

$$\eta_{\text{dream}} = \frac{\Delta \mathcal{P}_{\text{future}}}{Q_{\text{offline}}} \quad (28)$$

Dream gain per joule: improvement in future generalization/control divided by energy spent offline.

$$\mathcal{M} = \frac{I(Y_{n+1}; \Phi_n^{\text{phys}} | W_n, X_n)}{H(Y_{n+1} | W_n, X_n)} \quad (29)$$

Mortality index from equation (14), repeated here as a primary outcome.

12.2 Protocol 1 - Gibbs/CRR kernel calibration

1. Construct a small stochastic physical network with at least three stable or metastable output states a and independently measurable physical energies E_a .
2. Encode known archive weights ϕ_a and coherence labels C_a into slow physical parameters according to equation (9).
3. Measure the empirical state-selection distribution across repeated thermal trajectories at several temperatures/noise amplitudes.
4. Fit no free shape parameter after calibration; test whether $\log[P(a)/\phi_a]$ is linear in C_a with slope $1/\Omega_{\text{impl}}$ on held-out configurations.
5. Compare against ordinary Gibbs, softmax, and arbitrary learned-energy baselines. Failure of the predicted linear relation rejects this implementation mapping.

12.3 Protocol 2 - endogenous stopping-time test

6. Choose a stochastic task with natural variable completion times rather than an externally imposed fixed horizon.
7. Estimate $p(z, t)$ online and compute both C_l and C_A without using future data.
8. Train the stopping rule only on a training set; freeze all thresholds/hazard parameters.
9. On held-out runs compare stopping-time prediction, task loss, dissipation, and latency for CRR clocks, wall-clock, instantaneous loss, energy budget, and a flexible learned hazard.
10. A CRR clock is supported only if it adds predictive or efficiency value out of sample, not merely if it correlates with elapsed time.

12.4 Protocol 3 - mortality and transplant

11. Train two nominally identical analogue devices A and B independently on the same task distribution.
12. Copy all portable logical variables W_A into B while preserving B's own microscopic substrate state.
13. Measure the behavioural divergence between A and B and estimate M .
14. Ablate or randomize candidate archive variables Φ_A and test whether A loses the non-portable competence predicted by M .
15. Distill A's behaviour into a fresh device C without copying Φ_A . Compare recovery by distillation versus direct parameter copy.

12.5 Protocol 4 - reset and regeneration

16. Train a physical or simulated agent in recurring contexts, then trigger a controlled rupture/reset of fast state.
17. Compare fixed-prior reset, full checkpoint restoration, uniform archive replay, recency replay, and CRR reconstruction.
18. Measure G_R , recovery latency, context identification, interference, and energy.
19. Perturb archive content causally. CRR requires post-reset behaviour to change in the direction predicted by the perturbed archive weights.

12.6 Protocol 5 - wake/sleep schedule

20. Use a nonstationary continual-learning task with recurring contexts and a fixed total energy/compute budget.
21. Compare no offline phase, fixed-period offline phase, performance-triggered offline phase, and CRR-clock-triggered offline phase.
22. Within offline phases compare uniform replay, prioritized replay, and CRR weighting across multiple Ω_{off} values or annealing schedules.
23. Report old-task retention, new-task learning, return-to-context speed, calibration, and η_{dream} .
24. A “sleep” interpretation is unsupported if CRR scheduling provides no advantage after controlling total replay samples and energy.

12.7 Protocol 6 - world-model dreaming

25. Train a recurrent world model on real interaction, following a Dreamer-like separation between model learning and imagined trajectories.
26. During offline CRR phases reduce external input and generate trajectories from P_{off} in equation (20).
27. Partition offline computation into selective low- Ω and broad high- Ω phases; compare with a single fixed replay temperature.
28. Measure prediction generalization, counterfactual planning, rare-event coverage, hallucination/error accumulation, and subsequent real-world control.
29. Require improvement on held-out environmental regimes; self-consistency of generated dreams is not sufficient.

12.8 Protocol 7 - distributed event timing

30. Deploy heterogeneous mortal agents with local archives and sparse event communication as in equations (22)-(23).
31. Keep message content, bandwidth, and total message count fixed while experimentally shifting arrival phase relative to receiver C_i .
32. Measure collective reward, convergence, error propagation, communication energy, and phase order r .
33. Compare event-driven CRR communication with periodic synchronization and asynchronous content-only baselines.
34. Support for a CRR timing effect requires a reproducible phase-dependent performance curve beyond simple network latency.

12.9 Protocol 8 - thermodynamic accounting

35. Measure heat/work where experimentally feasible and estimate entropy production using the appropriate stochastic-thermodynamic model.
36. Quantify entropy erased at each reset rather than assuming one bit per rupture.
37. Plot performance and G_R against dissipation, not just wall-clock speed.
38. Use Manzano et al.-style stopping-time thermodynamics or an appropriate equivalent to ensure comparisons remain valid under stochastic computation time.
39. CRR is valuable thermodynamically only if its lifecycle control improves the Pareto frontier of performance, latency, or retained competence versus energy.

13. Predictions, falsifiers, and failure modes

Claim	Supporting observation	Falsifier / weakening result
-------	------------------------	------------------------------

Process clock matters	C predicts useful stopping better than elapsed time and instantaneous loss	No out-of-sample predictive gain after controlling simpler variables
Archive is computationally causal	Archive intervention changes post-reset reconstruction as predicted	Archive can be randomized with no effect once portable state is retained
Mortality is measurable	$M > 0$ and direct transplant loses competence while distillation can recover function	Portable state fully predicts future behaviour across substrates
Gibbs implementation is natural	Measured selection follows $\phi \exp(C/\Omega_{\text{impl}})$	Requires arbitrary nonphysical correction terms or unstable calibration
Offline regeneration helps	CRR-triggered/reweighted offline phases improve continual learning per joule	Ordinary replay matches or dominates at equal resources
Ω controls breadth	Changing effective temperature/noise produces predicted flattening/peaking of archive sampling	Sampling breadth does not track Ω_{impl} or harms reconstruction monotonically
Distributed timing matters	Phase-relative message delivery changes collective performance at fixed content	Only content/bandwidth/latency matter; receiver phase adds no explanatory power
Strong $C\Omega=1$ threshold	Independent physical events cluster near frozen CRR threshold	Threshold drifts by task, sampling scale, or hardware and learned hazard dominates

Several known conceptual hazards should be treated explicitly. Fisher path length can depend on the level of statistical description, and noisy sample paths require a well-defined probability model or coarse-graining. Arc length and squared-speed action cannot be conflated. Landauer cost cannot be inferred from a binary alphabet alone. Temperature/noise can improve or degrade computation non-monotonically, so Ω should not be assumed to be a universally beneficial “creativity knob.” Finally, sleep and phenomenological language can guide architecture without constituting evidence of machine experience.

14. Research programme and conclusion

CRR is most promising for thermodynamic and mortal computation when interpreted as a lifecycle heuristic with measurable state variables, not as a universal metaphysical law. Thermodynamic computing already provides physical stochastic sampling. Stochastic thermodynamics already provides dissipation and stopping-time theory. FEP already provides a broad principle of adaptive self-maintenance. World-model reinforcement learning already provides imagination. Sleep/replay research already provides mechanisms for offline memory processing. Mortal computation already provides the substrate-specificity thesis.

The distinctive CRR proposal is the composition of those pieces into a repeated physical lifecycle:

$$\text{world} \rightarrow C \rightarrow \delta \rightarrow \Phi \rightarrow R_{\Omega} \rightarrow \text{world} \quad (30)$$

In this interpretation, C is an endogenous process coordinate; δ is a commit or reconfiguration boundary; Φ is the physically embodied historical residue; R_{Ω} is the reconstruction process; and the next online regime is not reloaded from an abstract immortal checkpoint but rebuilt from the material history of this particular machine.

The direct Gibbs correspondence in equations (9)-(11) provides the strongest route from CRR mathematics to physical hardware. The mortality index in equation (14) provides a route from Hinton’s qualitative thesis to an empirical measurement. The wake/sleep architecture in equations (19)-(20) provides a route from biological replay and generative-model optimization to controllable offline phases. Equations (22)-(25) provide a route from one mortal device to a distributed ecology in which internal states remain local while completed acts become learning material for others.

If these ideas work, a CRR-native machine would differ from ordinary AI in a fundamental but testable way. Its function would depend not only on its current parameters and inputs but on the trajectory of its physical body. It would have endogenous computational epochs rather than merely an external training loop. It would periodically loosen sensory coupling and reconstruct its model from its own history. It would exchange sparse completed acts with other non-identical agents rather than assuming globally synchronized copyable state. Its “lifetime” would therefore be part of the computation.

Recommended first paper / experiment

Build or simulate a small stochastic physical learner with a fast state and slow physical archive. Compare fixed-time versus CRR-clock stopping, ordinary versus CRR-weighted reconstruction, and direct state transplant versus behavioural distillation under an equal energy budget. Compute both Fisher arc-length and Fisher-action clocks, M , G_R , η_C , and η_{dream} . A convincing result would not require the universal π/CV claims; it would require CRR lifecycle variables to predict and improve physical adaptive computation out of sample.

The deeper philosophical resonance can then be investigated without overclaiming. A machine whose present depends on retained physical history and anticipated future trajectories has an operational temporal structure that can be compared with phenomenology and process philosophy. But engineering success would not establish consciousness, and philosophical elegance would not substitute for energetic or behavioural evidence. The appropriate scientific sequence is therefore: build the clock, perturb the archive, measure the reconstruction, account for the heat, and only then ask how far the resulting form of temporality deserves to be interpreted.

References

- Chang, H., Tang, W., Wulf, A. M., et al. (2025). Sleep microstructure organizes memory replay. *Nature*, 637, 1161-1169. <https://doi.org/10.1038/s41586-024-08340-w>
- Georgescu, I. (2021). 60 years of Landauer's principle. *Nature Reviews Physics*, 3, 770.
- Hafner, D., Pasukonis, J., Ba, J., & Lillicrap, T. (2025). Mastering diverse control tasks through world models. *Nature*, 640, 647-653. <https://doi.org/10.1038/s41586-025-08744-2>
- Hinton, G. E. (2022). The Forward-Forward Algorithm: Some Preliminary Investigations. *arXiv:2212.13345*.
- Hinton, G. E., Dayan, P., Frey, B. J., & Neal, R. M. (1995). The wake-sleep algorithm for unsupervised neural networks. *Science*, 268(5214), 1158-1161.
- Hobson, J. A., Hong, C. C.-H., & Friston, K. J. (2014). Virtual reality and consciousness inference in dreaming. *Frontiers in Psychology*, 5, 1133. <https://doi.org/10.3389/fpsyg.2014.01133>
- Husserl, E. (1991). *On the Phenomenology of the Consciousness of Internal Time (1893-1917)*. Trans. J. B. Brough. Kluwer.
- Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3), 183-191.
- Manzano, G., Kardes, G., Roldan, E., & Wolpert, D. H. (2024). Thermodynamics of computations with absolute irreversibility, unidirectional transitions, and stochastic computation times. *Physical Review X*, 14, 021026. <https://doi.org/10.1103/PhysRevX.14.021026>
- Melanson, D., Abu Khater, M., Aifer, M., Donatella, K., Gordon, M. H., Ahle, T., Crooks, G., Martinez, A. J., Sbahi, F., et al. (2025). Thermodynamic computing system for AI applications. *Nature Communications*, 16, 3757. <https://doi.org/10.1038/s41467-025-59011-x>
- Ororbia, A., & Friston, K. (2023). Mortal Computation: A Foundation for Biomimetic Intelligence. *arXiv:2311.09589*.
- Sabine, A. (2026a). CRR: A Temporal Grammar. *temporalgrammar.ai*. Accessed 30 August 2026.
- Sabine, A. (2026b). The Paradox That Must Not Be Resolved: Transitional Space, the Relational Weave, and a Temporal Grammar for Finite Systems, v01. Working paper, August 2026.
- Spens, E., & Burgess, N. (2024). A generative model of memory construction and consolidation. *Nature Human Behaviour*, 8, 526-543. <https://doi.org/10.1038/s41562-023-01799-z>
- Watson, R. A., & Levin, M. (2023). The collective intelligence of evolution and development. *Collective Intelligence*, 2(2). <https://doi.org/10.1177/26339137231168355>
- Whitelam, S., & Casert, C. (2026). Nonlinear thermodynamic computing out of equilibrium. *Nature Communications*, 17, 1189. <https://doi.org/10.1038/s41467-025-67958-0>
- Wolpert, D. H., & Korbelt, J. (2024). Is stochastic thermodynamics the key to understanding the energy costs of computation? *Proceedings of the National Academy of Sciences*, 121(45), e2321112121.
- Zahavi, D. (2025). Edmund Husserl. *Stanford Encyclopedia of Philosophy* (current online entry consulted 30 August 2026).

Appendix A. Mathematical notation and implementation notes

A.1 Statistical state estimation

The Fisher clock requires a probability model, not a raw noisy sample path. In hardware, $p(z,t)$ may be estimated from repeated trajectories, local ensembles, a parametric state estimator, or a calibrated latent probabilistic model. The chosen description must be fixed before testing stopping predictions; changing coarse-graining after observing events invalidates the test.

A.2 Arc length versus action

For each run compute both v_{FR} , $C_I = \int v_{FR} dt$, and $C_A = \int v_{FR}^2 dt$. Report sampling frequency, estimator smoothing, and confidence intervals. The clocks should not be rescaled separately to make the same event threshold fit. A strong comparison freezes calibration on training hardware and evaluates on held-out runs or devices.

A.3 Archive measurement

The archive should be physically operational rather than a residual label for all unmodelled state. Candidate archive variables must be identified, measured, and intervened upon. The strongest evidence for regeneration is causal: targeted modification of Φ changes the distribution of post-reset states in the predicted direction while contemporaneous inputs are controlled.

A.4 Effective Ω

For a Gibbs implementation, $\Omega_{impl} = kBT/\alpha$ is directly calibratable. This is an implementation parameter and should not be conflated automatically with the stronger CRR claim that Ω is fixed by topological geodesic extent. The latter can be tested only after the physical state manifold and its symmetry have been independently justified.

A.5 Energy accounting

Report actual dissipated energy where measurable, not only theoretical Landauer minima. Separate online inference energy, reset energy, archive-writing energy, and offline regeneration energy. For stochastic stopping, use a thermodynamic framework valid for variable computation times.

A.6 Machine phenomenology

Terms such as retention, protention, sleep, dream, and mortality are used at two levels. At the engineering level they denote measurable operations: persistent history, predicted futures, reduced sensory coupling, internally generated trajectories, and substrate dependence. Any claim of phenomenal experience would require independent arguments and evidence and is outside the scope of this paper.